

ORIGINAL ARTICLE



Arrhythmias Other Than Atrial Fibrillation in Those With an Irregular Pulse Detected With a Smartwatch

Findings From the Apple Heart Study

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BACKGROUND: The Apple watch irregular pulse detection algorithm was found to have a positive predictive value of 0.84 for identification of atrial fibrillation (AF). We sought to describe the prevalence of arrhythmias other than AF in those with an irregular pulse detected on a smartwatch.

METHODS: The Apple Heart Study investigated a smartwatch-based irregular pulse notification algorithm to identify AF. For this secondary analysis, we analyzed participants who received an ambulatory ECG patch after index irregular pulse notification. We excluded participants with AF identified on ECG patch and described the prevalence of other arrhythmias on the remaining participant ECG patches. We also reported the proportion of participants self-reporting subsequent AF diagnosis.

RESULTS: Among 419 297 participants enrolled in the Apple Heart Study, 450 participant ECG patches were analyzed, with no AF on 297 ECG patches (66%). Non-AF arrhythmias (excluding supraventricular tachycardias <30 beats and pauses <3 seconds) were detected in 119 participants (40.1%) with ECG patches without AF. The most common arrhythmias were frequent PACs (burden ≥1% to <5%, 15.8%; ≥5% to <15%, 8.8%), atrial tachycardia (≥30 beats, 5.4%), frequent PVCs (burden ≥1% to <5%, 6.1%; ≥5% to <15%, 2.7%), and nonsustained ventricular tachycardia (4–7 beats, 6.4%; ≥8 beats, 3.7%). Of 249 participants with no AF detected on ECG patch and patient-reported data available, 76 participants (30.5%) reported subsequent AF diagnosis.

CONCLUSIONS: In participants with an irregular pulse notification on the Apple Watch and no AF observed on ECG patch, atrial and ventricular arrhythmias, mostly PACs and PVCs, were detected in 40% of participants. Defining optimal care for patients with detection of incidental arrhythmias other than AF is important as AF detection is further investigated, implemented, and refined.

GRAPHIC ABSTRACT: An online [graphic abstract](#) is available for this article.

Key Words: algorithms ■ atrial fibrillation ■ humans ■ pragmatic clinical trials as topic ■ prevalence

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WHAT IS KNOWN?

- The Apple watch irregular pulse detection algorithm was found to have a positive predictive value of 0.84 for identification of atrial fibrillation (AF).
- The prevalence of arrhythmias other than AF in those with an irregular pulse detected on a smartwatch is unknown.

WHAT THE STUDY ADDS

- In patients without AF detected on the ECG patches in the Apple Heart Study, 40% had arrhythmias other than short supraventricular tachycardias or pauses, with atrial and ventricular ectopy most commonly detected.
- Almost a third of participants with no AF detected on ECG patch reported they were diagnosed with AF by the end of the study.
- In parallel to refining approaches for early identification of AF patients who would benefit from treatment, defining optimal care for patients with detection of incidental arrhythmias other than AF is needed.

Nonstandard Abbreviations and Acronyms

AF	atrial fibrillation
ASD	absolute standardized difference
SVT	supraventricular tachycardia
VT	ventricular tachycardia

The Apple Heart Study characterized the ability of a smartwatch to correctly identify rhythms consistent with atrial fibrillation (AF) by sampling pulse rate data and analyzing irregularity.¹ Assessment of pulse irregularity for detection of AF is susceptible to arrhythmias other than AF,¹ many of which have unclear clinical significance and less well-defined care pathways.² Similarly, direct rhythm monitoring frequently captures incidental arrhythmias other than AF.² For patients and healthy users, detection of arrhythmias other than AF by both direct-to-consumer technology and traditional heart rhythm monitors may result in additional (or unnecessary) testing and associated anxiety.³

AF detection continues to be investigated,^{3,4} implemented,⁵ and made available direct to consumer,⁶ with the potential for intervention before the development of AF sequelae, such as stroke.⁵ As part of these efforts, increasing numbers of patients with rhythms other than AF will be identified. We, therefore, sought to describe arrhythmias other than AF detected in the Apple Heart Study on ECG patches, at any time and concordant with pulse irregularity detected by the Apple Watch. Additionally, we described the proportion of patients with non-AF

rhythms on initial ECG patch who reported a subsequent AF diagnosis.

METHODS

Study Design

Sponsored by Apple, the Apple Heart Study was a prospective, single-group, open-label, siteless, pragmatic study conducted by the Stanford Center for Clinical Research, the Stanford Center for Digital Health, and the Stanford Quantitative Sciences Unit. Study methods and initial findings have been previously described in detail.^{1,7} The data is not available to other researchers for purposes of reproducing the results.

A siteless study design was made feasible by a study app that verified eligibility, obtained participants' consent, provided study education, and directed participants through the study procedures. With a direct-to-participant enrollment strategy, the study app was available for download from the Apple App Store from November 29, 2017, to August 1, 2018. Major study eligibility criteria included (1) possession of a compatible Apple iPhone and Apple Watch, (2) ≥ 22 years of age, (3) US residency, and (4) proficiency in English, as reported by the participant. Study exclusion criteria included history of AF or current use of an oral anticoagulant.

Data Collection

After participants provided consent, the irregular pulse notification algorithm was activated, which leveraged the smartwatch's photoplethysmography sensor to intermittently measure pulse intervals (tachograms) over 1 minute while the participant was at rest. Based on variation in the pulse interval, the app classified these tachograms as regular or irregular, with recurrent irregular tachograms resulting in the participant receiving an irregular pulse notification through the watch. Additional details on the irregular pulse notification algorithm are available in Methods in the [Data Supplement](#) and Figure I in the [Data Supplement](#). After an irregular pulse notification, the study app continued to record and analyze tachograms that were time synchronized to the ECG strips collected on a subsequently worn ECG patch.

The irregular pulse notification prompted participants to initiate a telemedicine study visit directly from the app, which was conducted by a national telehealth servicer (American Well) and included (1) urgent symptom assessment and referral to urgent care or emergency department if indicated, (2) confirmation of study eligibility, and (3) for those eligible without urgent symptoms, mailing an ECG patch to the participant to wear for up to 7 days and return by mail. Participants were then contacted for a second telemedicine study visit to discuss ECG patch results and were prompted by the study app to complete an in-app survey 90 days after irregular pulse notification and complete a web-based end-of-study survey.

Rhythm Adjudication

The Stanford rhythm adjudication team included 14 board-certified physicians. The ECG patch reports were generated by trained technicians from the manufacturer (BioTelemetry), with the same process used for routine clinical care. The reports included structured data fields on ectopy and arrhythmia counts, rhythm strips of interest, and a summary statement

on overall findings. Atrial tachycardias were reported as supra-ventricular tachycardias (SVTs) on these reports. Two members of the Stanford rhythm adjudication team reviewed each report electronically for rhythms of interest (Table I in the [Data Supplement](#)), which included AF or atrial flutter exceeding 30 seconds in duration, nonsustained and sustained ventricular tachycardia (VT) or fibrillation, and pathological atrioventricular block. Discrepant interpretations between two members were reconciled by consensus of a committee of physicians. All other rhythms and rhythm counts were directly extracted from the ECG report.

Separately, 3-minute ECG strips time aligned to the sampled tachograms were reviewed by the adjudicators for rhythms of interest (Table I in the [Data Supplement](#)) through a custom REDCap interface/database. All 3-minute ECG strips were reviewed independently by 2 physicians, with disagreements resolved by a third clinician and then a committee, if necessary.

Analysis Cohorts

The participants included in this study were those who received an irregular pulse notification and provided data on an ECG patch. More specifically, we excluded those (1) who did not receive an irregular pulse notification, (2) who did not initiate a first study visit after irregular pulse notification, (3) who at first study visit reported urgent symptoms or main study exclusion criteria (eg, prior AF or anticoagulant use), (4) who did not return their ECG patch or returned an unreadable ECG patch, and (5) who had delays in application or return of ECG patch. For the included participants, tachograms (both individual and clusters that would have prompted irregular pulse notifications) occurring while the participants wore the ECG patch were sampled for analysis to prevent overrepresentation of tachograms from participants with frequent irregularity. Sampling methodology is outlined in the [Data Supplement](#).

Statistical Analyses

The primary analysis of this substudy was to describe arrhythmias from the ECG patch report in participants with prior irregular pulse notification without AF detected on the ECG patch (Table I in the [Data Supplement](#)). We determined the proportion of patients with any arrhythmia detected, excluding (1) pauses 2 to 3 seconds and (2) SVT 3 to 9 beats or 10 to 29 beats, which were not clearly pathological and common, respectively. Secondary analyses were to describe non-AF rhythms captured on the ECG patch that time synchronized to (1) irregular tachograms and (2) clusters of tachograms that met criteria for an irregular pulse notification. Notably, after the initial pulse notification that triggered ECG patch receipt, pulse irregularity measurements by the watch-based algorithm continued allowing for these secondary analyses, although the patient received no subsequent irregular pulse notifications. We compared baseline characteristics (1) between those with and without AF detected on the ECG patch and (2) for those without AF, between those ≥ 65 and < 65 years of age. We compared arrhythmia frequencies between participants ≥ 65 and < 65 years of age. We also determined the proportion of participants who reported being told by a doctor they had AF on either end-of-study or 90-day postnotification surveys, both for all included participants and those with an arrhythmia other than SVT < 30 beats or pauses 2 to 3 seconds. Urgent arrhythmias

(Table I in the [Data Supplement](#)) detected on any ECG patch, regardless of exclusion criteria, were also described.

Means and SDs are provided for continuous characteristics, and frequency distributions with percentages are presented for binary and categorical characteristics. Differences in arrhythmia frequencies by groups of interest were assessed using absolute standardized differences (ASD). The ASD, expressed in units of standard deviations, is interpreted using the Cohen d statistic, with the following guidelines for interpretation ($d=0.2$, small difference; $d=0.5$, medium difference; $d=0.8$, large difference; $d<0.2$, trivial difference).⁸ We presented ASD in place of P as our goal was to describe imbalance between the observed groups rather than test hypotheses.⁹

The analyses presented here were performed by Stanford data scientists independent of the sponsor. Analyses were performed using R (3.5.1).¹⁰ The principal investigators (M.V.P., M.P.T., M.D.) and study chair (K.W.M.) had full access to all study data and take responsibility for its integrity and the data analysis. The research protocol was approved by the institutional review board at the Stanford University and by a central institutional review board (Advarra).

RESULTS

The Apple Heart Study cohort included 419 297 participants, of whom 2161 participants (0.52%) received an irregular pulse notification. From the notified participants, 450 participants returned an analyzable ECG patch after participating in a study visit and 297 (66%) of these had no AF detected (Figure 1). Participants with an ECG patch without AF detected, as compared with those with AF detected, were somewhat younger (57.7 ± 14.8 versus 60.3 ± 11.3 ; ASD, 0.202) and more likely to be female (25.6% versus 17.0%; ASD, 0.237; Table 1). ECG patch wear times and interval from notification to patch application were similar between participants without and with AF on ECG patch (wear time, 6.7 days [25th to 75th percentile, 6.5–6.9] versus 6.8 days [25th to 75th percentile, 6.6–6.9]; notification interval, 7.2 days [25th to 75th percentile, 4.8–13.3] versus 8.3 days [25th to 75th percentile, 5.9–13.5]).

Non-AF Rhythms on the Full ECG Patch

For the 297 participants without AF detected on ECG patch, any arrhythmia (excluding SVT < 30 beats and pauses < 3 seconds) was detected in 119 participants (40.1%). Atrial ectopy burden of 1% to 5%, 5% to 15%, and $\geq 15\%$ was seen in 47 (15.8%), 26 (8.8%), and 4 (1.3%) of the 297 participants, respectively. SVT ≥ 30 beats occurred in 16 participants (5.4%). Ventricular ectopy burden of 1% to 5%, 5% to 15%, and $\geq 15\%$ was seen in 18 (6.1%), 8 (2.7%), and 2 (0.7%) of the 297 participants, respectively. Short and moderate length nonsustained VTs were not uncommon (4–7 beats, 19 participants [6.4%]; ≥ 8 beats, 11 participants [3.7%]), while sustained VT was not detected in any participant. Heart block (Mobitz II, high grade, or complete) and pauses ≥ 3 seconds were detected in 3 (1.0%) and 1

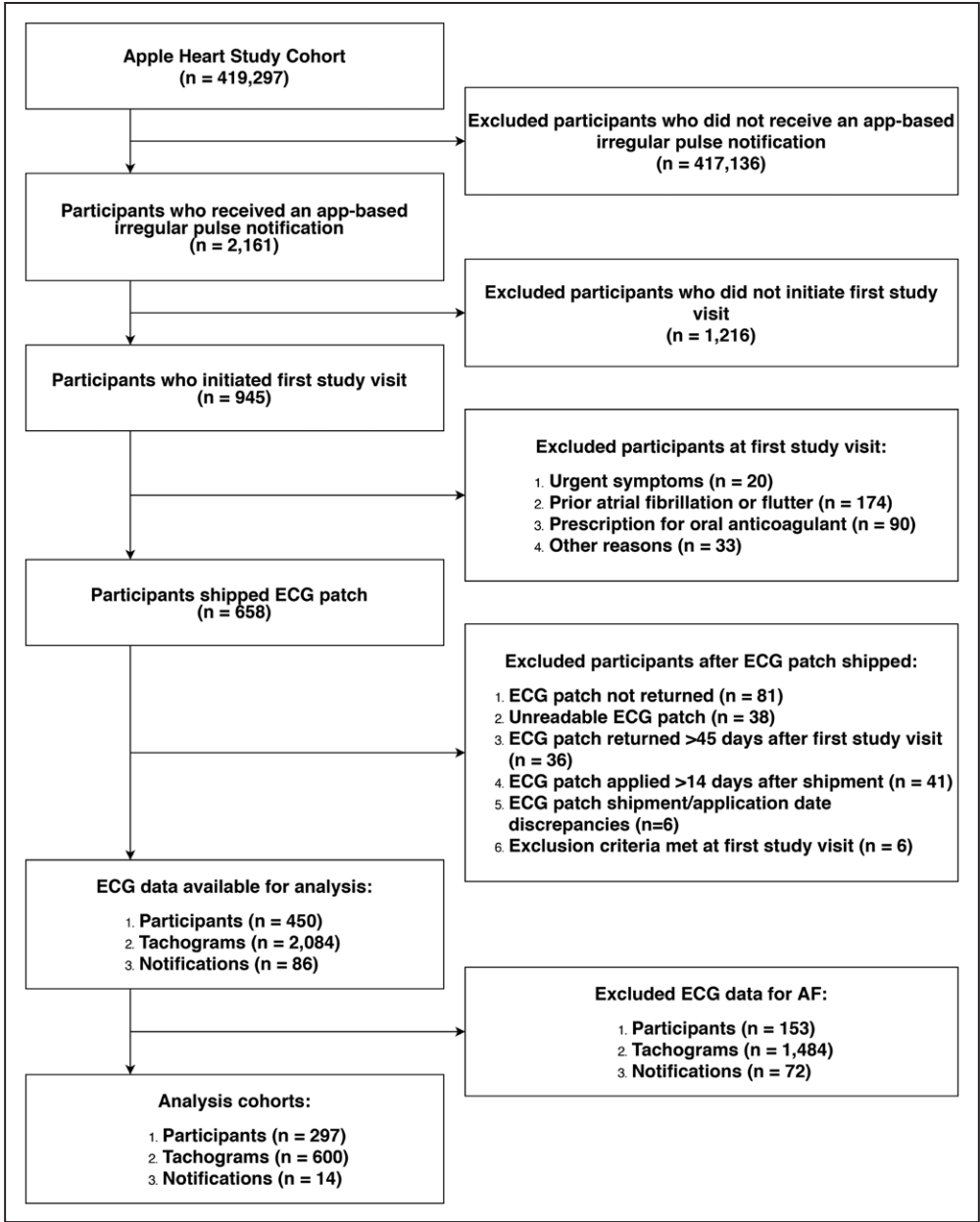


Figure 1. Cohort selection diagram. Inclusion and exclusion criteria used to select analysis cohorts. See methods in the [Data Supplement](#) for tachogram sampling strategy. AF indicates atrial fibrillation. App indicates application.

(0.3%) participants, respectively. SVT of 3 to 9 and 10 to 29 beats was detected in 108 (36.4%) and 79 (26.6%), respectively (Table 2).

Participants without AF detected ≥ 65 years of age, as compared with those < 65 years, were more likely to be female (32.3% versus 21.2%; ASD, 0.409) and White (92.4% versus 80.4%; ASD, 0.521), with higher rates of most comorbidities (Table 1). Atrial and ventricular ectopy and nonsustained VT were more frequently detected in participants without AF ≥ 65 years of age, as compared with those < 65 years of age (Table 2).

Patient-reported data on AF status were available for 249 participants without AF detected on ECG patch

(83.8%), 76 of whom reported being told they had AF (30.5%) on either the end-of-study or 90-day postnotification survey. Among the 100 participants with an arrhythmia (excluding SVT < 30 beats and pauses < 3 seconds) and available data on patient-reported AF status, 36 reported being told they had AF (36.0%) on either survey.

Non-AF Rhythms Concordant With Irregular Tachograms

From the 450 readable ECG patch monitors, there were 2084 tachograms sampled, of which 600 did

Table 1. Baseline Characteristics

Characteristics	All patients with ECG patch (n=450)	AF on ECG patch (n=153)	No-AF on ECG patch*			ASD†‡	ASD†§
			No-AF all patients (n=297)	No-AF ≥65 y (n=118)	No-AF <65 y (n=179)		
Sex						0.237	0.409
Female	102 (22.7%)	26 (17.0%)	76 (25.6%)	38 (32.3%)	38 (21.2%)		
Male	335 (74.4%)	124 (81.0%)	211 (71.0%)	80 (67.8%)	131 (73.2%)		
Other	0	0	0	0	0		
Not reported	13 (2.9%)	3 (2.0%)	10 (3.4%)	0	10 (5.6%)		
Age, y	58.6±13.8	60.3±11.3	57.7±14.8	71.5±5.4	48.6±11.7	0.202	2.532
Age distribution, y						0.298	2.636
≥65	181 (40.2%)	63 (41.2%)	118 (39.7%)	118 (100.0%)	0		
55–64	114 (25.3%)	47 (30.7%)	67 (22.6%)	0	67 (37.4%)		
40–54	106 (23.6%)	34 (22.2%)	72 (24.2%)	0	72 (40.2%)		
22–39	49 (10.9%)	9 (5.9%)	40 (13.5%)	0	40 (22.3%)		
Race or ethnic group‡						0.211	0.521
White	379 (84.2%)	126 (82.4%)	253 (85.2%)	109 (92.4%)	144 (80.4%)		
Hispanic	20 (4.4%)	8 (5.2%)	12 (4.0%)	1 (0.8%)	11 (6.1%)		
Black	16 (3.6%)	7 (4.6%)	9 (3.0%)	2 (1.7%)	7 (3.9%)		
Asian	8 (1.8%)	4 (2.6%)	4 (1.3%)	2 (1.7%)	2 (1.1%)		
American Indian	3 (0.7%)	1 (0.7%)	2 (0.7%)	2 (1.7%)	0		
Pacific Islander	0	0	0	0	0		
Middle Eastern	2 (0.4%)	0	2 (0.7%)	0	2 (1.1%)		
Other or mixed race	6 (1.3%)	3 (2.0%)	3 (1.0%)	1 (0.8%)	2 (1.1%)		
Not reported	16 (3.6%)	4 (2.6%)	12 (4.0%)	1 (0.8%)	11 (6.1%)		
Medical history							
Obesity	192 (42.7%)	66 (43.1%)	126 (42.4%)	45 (38.1%)	81 (45.3%)	0.087	0.399
Hypertension	200 (44.4%)	63 (41.2%)	137 (46.1%)	74 (62.7%)	63 (35.2%)	0.144	0.635
Diabetes	53 (11.8%)	18 (11.8%)	35 (11.8%)	16 (13.6%)	19 (10.6%)	0.088	0.351
Heart failure	10 (2.2%)	7 (4.6%)	3 (1.0%)	2 (1.7%)	1 (0.6%)	0.233	0.36
Stroke or TIA	10 (2.2%)	2 (1.3%)	8 (2.7%)	5 (4.2%)	3 (1.7%)	0.134	0.374
Peripheral artery disease	10 (2.2%)	3 (2.0%)	7 (2.4%)	6 (5.1%)	1 (0.6%)	0.092	0.44
CHA ₂ DS ₂ -VASc score ≥2	171 (38.0%)	52 (34.0%)	119 (40.1%)	92 (78.0%)	27 (15.1%)	0.132	1.756
Current smoking	10 (2.2%)	1 (0.7%)	9 (3.0%)	1 (0.8%)	8 (4.5%)	0.177	0.227
Alcohol: ≥1 drink/wk	227 (50.4%)	84 (54.9%)	143 (48.1%)	61 (51.7%)	82 (45.8%)	0.135	0.118

Values are mean±SD or n (%). AF indicates atrial fibrillation; ASD, absolute standardized difference; and TIA, transient ischemic attack.
*Analysis cohort.
†A larger ASD corresponds to a larger difference between the groups (ie, 0.2, small difference; 0.5, medium difference; 0.8, large difference; d<0.2, trivial difference).
‡ASD between No-AF all patients and AF on ECG patch groups.
§ASD between No-AF ≥65 y and No-AF <65 y groups.

not have AF detected on time-synchronized ECG strips. Of these, 566 tachograms (94.3%) correlated with a rhythm other than sinus. Atrial and ventricular ectopy were the most frequent rhythms detected (frequent PACs [≥6 over 3 minutes], 472 [78.7%]; frequent PVCs [≥6 over 3 minutes], 96 [16.0%]). Sinus arrhythmia and Mobitz Type I (Wenckebach) only were observed in 28 tachograms (4.7%) and 5 tachograms (0.8%), respectively. Atrial and ventricular ectopy were more frequently present on tachograms of participants ≥65 years of age, as compared with those <65 years of age (Table 3).

Non-AF Rhythms Concordant With Irregular Pulse Notifications

From the 450 readable ECG patch monitors, there were 86 irregular pulse notifications that were sampled, of which 14 did not have a contributory tachogram that correlated with AF on time-synchronized ECG strips. Of these, 12 notifications (86%) were in response to atrial tachycardia (median duration of the longest atrial tachycardia episode, 8 beats [25th to 75th percentile, 6–234]), 1 notification (7%) was in response to sinus arrhythmia only, and 1 notification (7%) was in response to sinus rhythm only.

Table 2. Arrhythmias on ECG Patch in Those Without AF

Rhythm*	No-AF all patients (n=297)	No-AF ≥65 y (n=118)	No-AF <65 y (n=179)	ASD†‡
Ectopy ≥1%, SVT ≥30 beats, ventricular arrhythmias, heart block, pauses ≥3 s, and tachycardia ≥200 beats/min				
Atrial ectopy§				0.594
PAC ≥1% to <5%	47 (15.8%)	26 (22.0%)	21 (11.7%)	
PAC ≥5% to <15%	26 (8.8%)	20 (16.9%)	6 (3.4%)	
PAC ≥15%	4 (1.3%)	1 (0.8%)	3 (1.7%)	
SVT ≥30 beats	16 (5.4%)	7 (5.9%)	9 (5.0%)	0.040
Ventricular ectopy§				0.322
PVC ≥1% to <5%	18 (6.1%)	12 (10.2%)	6 (3.4%)	
PVC ≥5% to <15%	8 (2.7%)	4 (3.4%)	4 (2.2%)	
PVC ≥15%	2 (0.7%)	0	2 (1.1%)	
NSVT				0.389
NSVT 4–7 beats	19 (6.4%)	11 (9.3%)	8 (4.5%)	
NSVT ≥8 beats	11 (3.7%)	9 (7.6%)	2 (1.1%)	
Ventricular tachycardia	0	0	0	NA
Ventricular fibrillation	0	0	0	NA
Heart block				
Mobitz II	0	0	0	NA
High grade	3 (1.0%)	0	3 (1.7%)	NA
Complete	0	0	0	NA
Pauses ≥3 s	1 (0.3%)	0	1 (0.6%)	NA
Heart rate >200 beats/min	0	0	0	NA
SVT ≤30 beats and pauses ≤3 s				
SVT 3–9 beats	108 (36.4%)	44 (37.3%)	64 (35.8%)	0.992
SVT 10–29 beats	79 (26.6%)	56 (47.5%)	23 (12.8%)	
Pauses ≥2 (<3) s	26 (8.8%)	8 (6.8%)	18 (10.1%)	0.118

Values are n (%). All values obtained from ePatch structured fields or narrative summary unless otherwise specified. AF indicates atrial fibrillation; ASD, absolute standardized difference; NA, not applicable; NSVT, nonsustained ventricular tachycardia; PAC, premature atrial contraction; PVC, premature ventricular contraction; and SVT, supraventricular tachycardia.

*Rhythms are not mutually exclusive.

†A larger ASD corresponds to a larger difference between the groups (ie, 0.2, small difference; 0.5, medium difference; 0.8, large difference; <0.2, trivial difference).

‡ASD between No-AF ≥65 y and No-AF <65 y groups.

§ASD calculation included patients with 0% to <1% ectopy.

||Arrhythmias identified by cardiologist overread of ePatch report.

Urgent Rhythms on ECG Patch

From the 539 readable ECG patches, which included patches regardless of whether they ultimately detected AF or were excluded from the final analysis cohort, 20 participants had an arrhythmia detected that met urgent criteria (Figure 1). These arrhythmias included AF with ventricular rates >200 beats per minute for >30 seconds (n=18), pause >6 seconds (n=1), and nonsustained VT >6 seconds (n=1; Figure 2).

DISCUSSION

We found that in patients without AF detected on the ECG patches in the Apple Heart Study, 40% had arrhythmias other than short SVTs or pauses, with atrial and ventricular ectopy most commonly detected.

Almost a third of participants with no AF detected on ECG patch reported they were diagnosed with AF by the end of the study. In parallel to refining approaches for early identification of AF patients who would benefit from treatment, defining optimal care for patients with detection of incidental arrhythmias other than AF is needed.

Although arrhythmias other than AF were frequently detected in those with a prior irregular pulse notification and no AF while wearing an ECG patch, it is uncertain whether detection of these rhythms reflected (1) patients with AF at the time of irregular pulse notification but without AF during the ECG patch wear time, with other arrhythmias incidentally identified; (2) arrhythmias identified by the ECG patch being analogous to the arrhythmias that triggered the original irregular pulse notification; or (3)

Table 3. Non-AF Arrhythmias on ECG Patch When Tachograms Were Irregular

Rhythm*	All tachograms (n=600)	≥65 y (n=366)	<65 y (n=234)	ASD†‡
Any PAC (≥1 over 3 min)	495 (82.5%)	316 (86.3%)	179 (76.5%)	0.255
Frequent PACs (≥6 over 3 min)	472 (78.7%)	303 (82.8%)	169 (72.2%)	0.255
Atrial tachycardia (≥3 beats)	204 (34.0%)	129 (35.2%)	75 (32.1%)	0.076
Atrial tachycardia (≥30 beats)	26 (4.3%)	21 (5.7%)	5 (2.1%)	0.189
Any PVC (≥1 over 3 min)	222 (37.0%)	146 (39.9%)	76 (32.5%)	0.155
Frequent PVCs (≥6 PVCs over 3 min)	96 (16.0%)	64 (17.5%)	32 (13.7%)	0.105
Ventricular tachycardia	0 (0%)	0 (0%)	0 (0%)	NA
Sinus arrhythmia	28 (4.7%)	17 (4.6%)	11 (4.7%)	0.136
Junctional beats	5 (0.8%)	3 (0.8%)	2 (0.9%)	0.004
Heart block				
Mobitz type I	5 (0.8%)	1 (0.3%)	4 (1.7%)	0.145
Mobitz type II or complete	0 (0%)	0 (0%)	0 (0%)	NA
Any nonsinus rhythm	566 (94.3%)	355 (97.0%)	211 (90.2%)	0.281

Values are n (%). AF indicates atrial fibrillation; ASD, absolute standardized difference; NA, not applicable; PAC, premature atrial contraction; and PVC, premature ventricular contraction.
*Rhythms are not mutually exclusive.
†A larger ASD corresponds to a larger difference between the groups (ie, d=0.2, small difference; d=0.5, medium difference; d=0.8, large difference; d<0.2, trivial difference).
‡ASD between No-AF ≥65 y and No-AF <65 y groups.

expected arrhythmias identified by ambulatory ECG monitoring in patients who meet Apple Heart Study inclusion criteria.

It is also uncertain whether detection of these arrhythmias, independent of a potential subsequent diagnosis of AF, will improve long-term outcomes. Frequent PACs and SVTs, the most common non-AF arrhythmias observed, are predictors for incident AF and ischemic stroke, independent of AF.^{11,12} However, available evidence does not suggest improvement in outcomes if these rhythms are treated or therapies aimed at reducing ischemic stroke risk are initiated.² For some participants, observed burden of PVCs exceeded thresholds previously associated with incident cardiomyopathy.¹³ However, benefit of PVC treatment is largely restricted to patients with symptoms or cardiomyopathy,¹³ with no evidence for routine screening of asymptomatic individuals. It is unknown whether the natural history and treatment benefit for these rhythms differs between patients included in prior studies and adopters of digital health technology, who may have different ages, comorbidity profiles, and symptom burdens.

Ectopy and other non-AF arrhythmias identified in our study can be sequelae of an underlying cardiomyopathy or other health condition, such as thyrotoxicosis.^{14,15} Importantly, arrhythmias may be the presenting sign of cardiomyopathy, occurring before congestive symptoms or identifiable structural abnormalities with conventional imaging.¹⁶ Consideration of these possibilities is prudent while evidence generation is ongoing for optimal care of patients with incidental arrhythmias other than AF detected by direct-to-consumer products.

Despite the participants included in this substudy having no AF detected on ECG patch, almost a third reported being diagnosed with AF on survey, collected 90 days after irregular pulse notification or at the end of study. For participants with any arrhythmia (other than short SVTs or pauses), the proportion reporting eventual AF diagnosis exceeded one-third. It is plausible that participants were initially misclassified as No-AF due to a low burden of paroxysmal AF. Sequential episodic or long-term rhythm monitoring may be beneficial in this scenario, which has been shown to substantially increase AF diagnostic yield when clinical suspicion for AF is high.¹⁷

As AF detection using pulse-based algorithms continues to be investigated, implemented, and made available directly to consumers, improving discrimination of AF from other arrhythmias through irregular pulse analysis is paramount. While older populations are more likely to have AF identified, they are also more susceptible to frequent PACs, which may explain why the positive predictive value of AF from an irregular pulse notification was lower in the ≥65 age group.¹ Contrary to what was anticipated, the proportion of participants with sinus arrhythmia was low and did not differ significantly across age groups. To define optimal care for those with arrhythmias other than AF identified by irregular pulse analysis, investigation should be prioritized into arrhythmia natural history, evaluation and management strategies, and value of ongoing intensive screening for AF.

Limitations

Study limitations included both participant dropout and ECG patch data only from those who received an

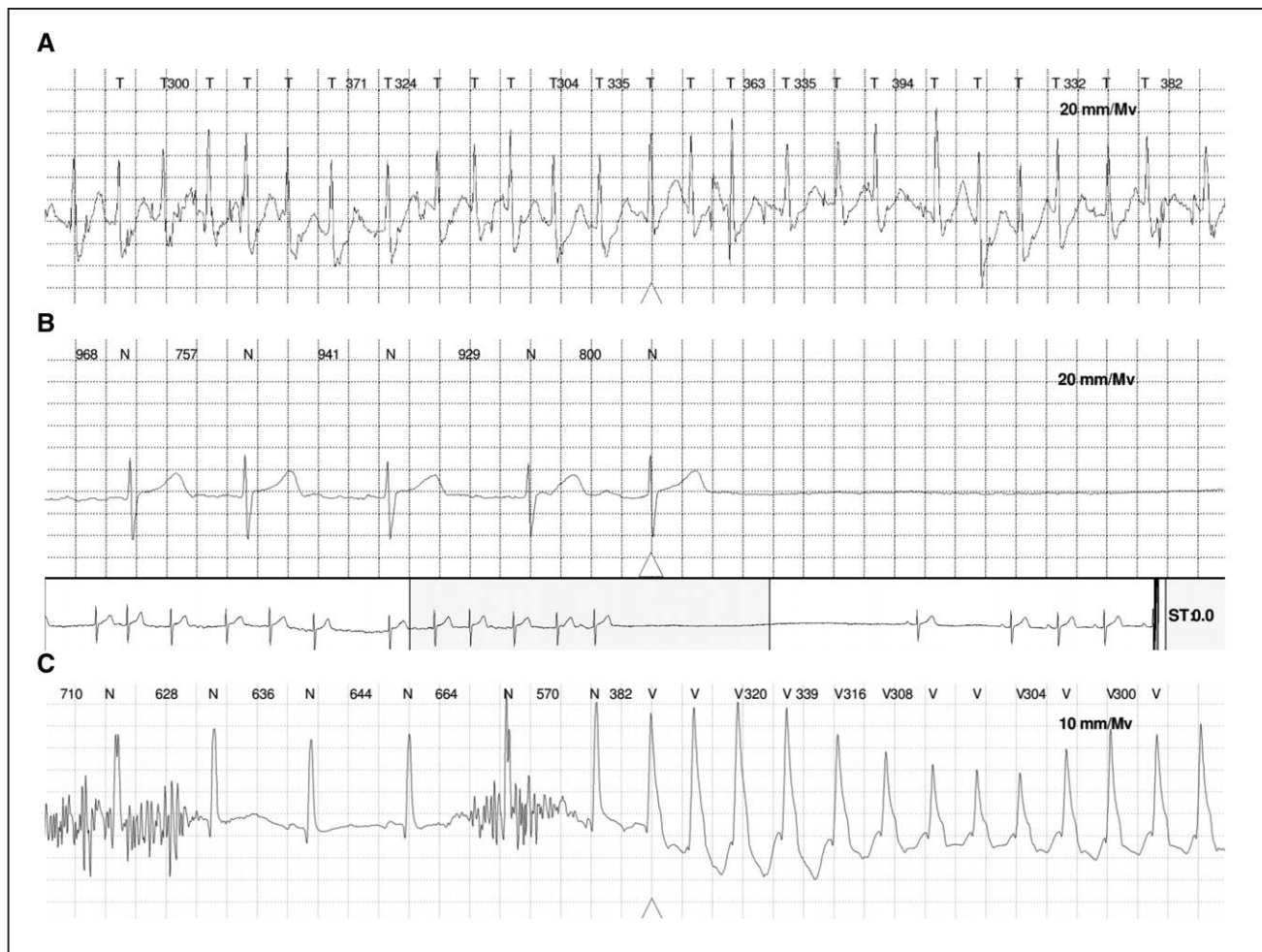


Figure 2. Urgent arrhythmias detected on study ECG patches.

A, A 73-y-old man in atrial fibrillation (AF) with ventricular rates >200 beats/min for >30 s. Patient referred to cardiology and initiated on rivaroxaban and amiodarone. **B**, A 55-y-old man with paroxysmal AF with conversion pauses up to 6.9 s. Patient referred to emergency room with inpatient cardiology evaluation. **C**, A 68-y-old man with wide complex tachycardia (longest 19 beats for 6.2 s [184 beats/min]). Patient referred to electrophysiology for evaluation.

initial notification. Coupled with the study inclusion criteria as outlined in the Methods, the results may not generalize to all adopters of digital health technology or the general population. Additionally, with multiple direct-to-consumer rhythm detection technologies available and in the pipeline, with rapid intraproduct iteration, issues with incidental detection of arrhythmias other than AF may not be uniform or static. The pragmatic study design specified ascertainment of outcomes reported by the participants, limiting the ability to confirm AF diagnosis, health care utilization, and other outcomes. Analyses performed at the tachogram and notification level did not account for multiple observations from the same participant; however, the tachogram sampling was designed to prevent overrepresentation of any patient's data. The study did not ascertain participants' symptoms while wearing ECG patches or long-term cardiovascular end points, preventing estimation of the arrhythmias' clinical significance. Importantly, only 14 patients who met criteria

for an irregular pulse notification while wearing the ECG patch did not have AF, providing a small sample size for analysis. Finally, the ECG patch's single-lead design, coupled with study design utilizing a 3-minute rhythm strip for tachogram adjudication, can make a precise arrhythmia diagnosis challenging, particularly the distinction between atrial tachycardia, frequent PACs, and sinus arrhythmia.

Conclusions

In the absence of AF, 40% of the Apple Heart Study participants with an irregular pulse notification who wore an ECG patch had an arrhythmia other than short SVT or pauses, which were predominantly atrial and ventricular ectopy. Notably, over a third of these participants reported they were diagnosed with AF by the end of the study. An irregular pulse notification that did not correspond to AF was most frequently attributable to atrial tachycardia. Defining optimal care for patients

with detection of incidental arrhythmias other than AF is needed as AF detection is further investigated and implemented.

ARTICLE INFORMATION

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Supplemental Materials

Apple Heart Study Investigators
Expanded Methods
Supplemental Table I
Supplemental Figure I

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